Standard operating procedure for *Isognomon alatus* search

Drs. James Douglass and Melissa May, Florida Gulf Coast University

Background: The Flat Tree Oyster, *Isognomon alatus*, is native to the tropical Western Atlantic, and is abundant in the Florida Keys and Southeast Florida (Benthotage et al., 2020; Bieler & Mikkelsen, 2004; Mikkelsen & Bieler, 2021). It is commonly associated with the roots of red mangroves (*Rhizophora mangle*) but can occur on other substrates, as well, in both intertidal and subtidal zones (Mikkelsen & Bieler, 2021; Wilk & Bieler, 2009). The geographic range boundary of *Isognomon alatus* is not well characterized, but anecdotal evidence suggests that it may be expanding into higher latitudes and/or becoming increasingly prevalent in higher latitude portions of its historic range. This pattern of change in distribution and abundance, known as "tropicalization," has been observed for many other species in Florida and globally, and can be ecologically and economically consequential (Vergés et al., 2014; Wernberg et al., 2024). As a primary and secondary space occupier, filter feeder, and biogenic structure-forming species, *Isognomon alatus* could interact strongly with established species such as Eastern Oysters (*Crassostrea virginica*), for better or worse. Clarifying the past and present distribution and abundance of *Isognomon alatus* would fill important gaps in our understanding of the ecology of this species and how that ecology might change in a tropicalization scenario.

Objectives:

1. To characterize the present abundance and distribution of *Isognomon alatus* through a simple, standardized search protocol replicated at numerous, discrete locations.
2. To analyze and report on the search's findings, ultimately producing a peer-reviewed manuscript for publication, co-authored by major contributors to the study and acknowledging all contributors.

Methods:

1. Standardized search using **Rapid Oyster Community Assessment Datasheet**
    a. This protocol is designed as a rapid characterization of epibiont species composition and cover on hard substrates in intertidal or shallow subtidal areas that might support *Isognomon alatus*.
    b. The first step is to select a manageable number of sites in your area to search. Consider selecting sites to span a range of hydrologic conditions, e.g., high salinity areas near inlets as well as lower salinity sites. It is recommended that you send a map or coordinates list of the proposed sites to the FGCU project organization team (jdouglass@fgcu.edu) to help keep track of the areas covered and identify gaps.
    c. Each site will be defined by a single, center point (Latitude and Longitude) but presumed to include all the searchable substrate in the general vicinity of the point; an area of roughly 50 m radius (Figure 1). **It is OK if sites vary in size** because abundance data will be recorded in a form that is inherently normalized to the area searched.

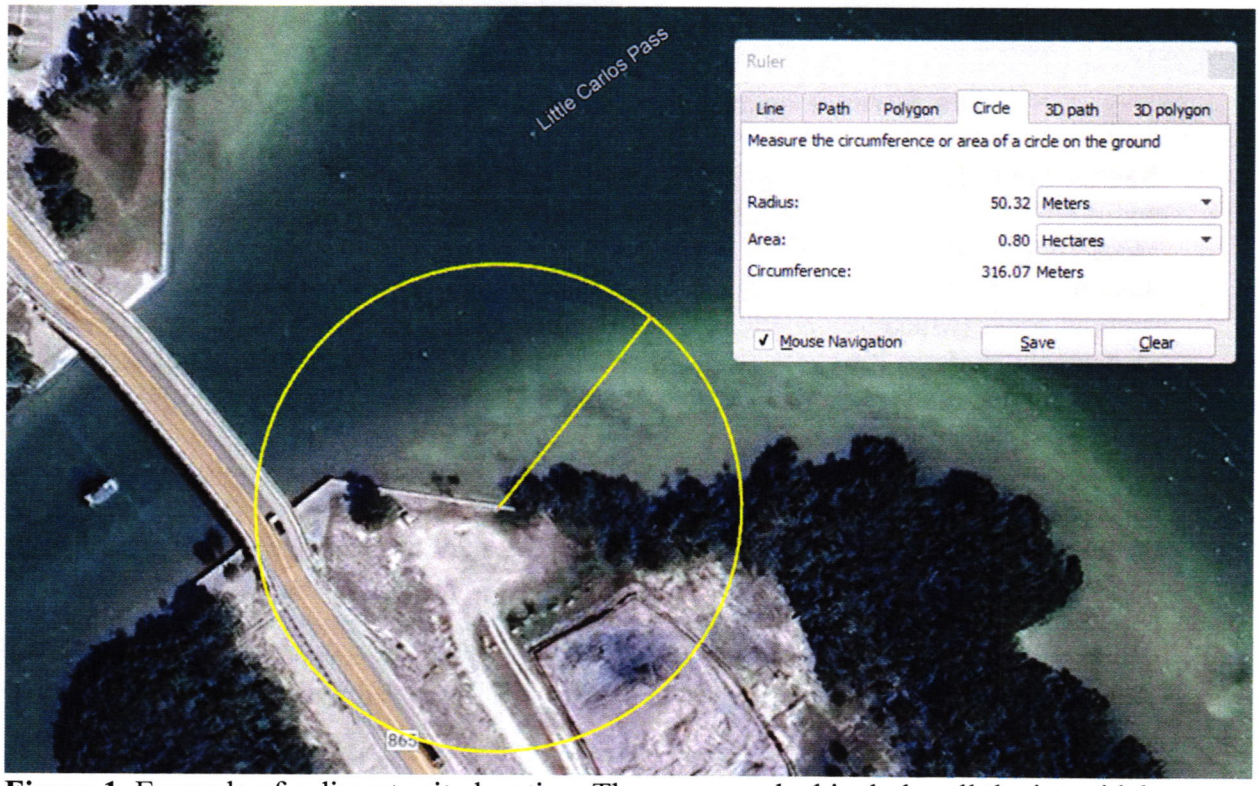


Figure 1. Example of a discrete site location. The area searched includes all the intertidal and shallow subtidal hard substrates within this area, e.g., the seawall, *Crassostrea* shell piles near the seawall, and roots of the red mangroves.

d. "**Suitable substrate**" includes any hard substrate in the intertidal or shallow subtidal zone of the search area that might support sessile epibiota. A single site may include multiple types of hard substrate (e.g., seawall, *Crassostrea virginica* oyster reef, mangrove roots.). All types of hard substrate searched at a site should be indicated on the datasheet. **You do NOT need to make separate epibiont abundance estimates for each substrate type,** but you're welcome to note substrate associations in the margins of the sheet, like, "*Isognomon* observed on both mangrove roots and seawall."

e. Epibiont abundance will be recorded using the **Braun-Blanquet scale** (Table 1); an ordinal scale of abundance based on the estimated proportion of an assessed area that is covered by a particular species or habitat type. Typically, Braun-Blanquet scores are assigned to small, well-defined search areas like quadrats, but in this case the Braun-Blanquet estimate will be for the site as a whole. E.g., if about 60% of the hard substrate at a site is covered in *Crassostrea virginica* and about 4% is covered in *Isognomon alatus*, you would give a BB score of "4" to *Crassostrea* and a BB score of "1" to *Isognomon.* **Circle the appropriate score value for each epibiont in the "BB Cover Category" column.**

f. For the intertidal and subtidal columns, mark a "1" if the species was present in that zone and a "0" if it was not. If you're only searching the intertidal zone (i.e., you're not snorkeling) mark "N/A" in the subtidal column.

Table 1. Key to Braun-Blanquet cover category system used in this survey. For some post-survey analyses and graphical presentations, BB Scores will be converted to equivalent percent cover values using the conversions in the right-hand column of this table.

BB Score	Description	% Cover equivalent
0.1	Very rare; only a few individuals encountered in a large search area	0.5%
0.5	Rare; sparsely present but with much less than 5% cover	1%
1	Common but covering less than 5% of the relevant substrate	3.5%
2	Any number of individuals covering 5-25% of the relevant substrate	15.5%
3	Any number of individuals covering 25-50% of the relevant substrate	38%
4	Any number of individuals covering 50-75% of the relevant substrate	63%
5	Any number of individuals covering >75% of the relevant substrate	88%

g. Some species and broader taxonomic categories of sessile epibiota common in Florida are already written into rows on the datasheet, but if other conspicuous epibionts are present they can be written into the blank rows. If a pre-written epibiont is not present at the site at all, just put a horizontal line through that row.
h. When your site surveys are completed, scan your datasheets and email them to jdouglass@fgcu.edu for entry into the database.

2. Data management, analysis, and reporting.
    a. The FGCU team will maintain a Microsoft Excel file (.xlsx format) and ArcGIS data layer (.shp format) for the *Isognomon* search data and will provide updated copies of the digital data to all study participants periodically and by request.
    b. The study will move from the data collection phase to the analysis and reporting phase when sufficient coverage of the study region is achieved, i.e., when all eight Florida Oyster Integrated Mapping and Monitoring Program regions of focus (Radabaugh et al. 2019) are represented by at least five survey sites and there are no gaps > 100 km between survey sites.
    c. Key figures and statistics for the final report / publication will include:
        i. A comprehensive map including all sites surveyed, with symbology indicating prevalence of *Isognomon alatus*, *Crassostrea virginica*, and other key epibiota.
        ii. Stacked column graphs summarizing mean percent cover equivalents of each epibiont assessed in each OIMMP region.
        iii. A multivariate statistical comparison of epibiont composition by site and region, and possibly by additional within-region factors (e.g., salinity regime, substrate, etc.).

3. Group coordination and authorship.
    a. We are envision a many-authored paper including the FGCU principle investigators as first or last authors, and those leading regional survey efforts or contributing in other significant ways as additional authors.

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